

INTEGRATION AND DEPLOYMENT OF A MULTI-MODEL AGRICULTURAL INTELLIGENCE SYSTEM

Abstract

This report presents a Smart Agriculture Decision Support System that integrates crop recommendation, soil profile segmentation, and crop yield-index prediction into a unified Python application. The system

Introduction

uses a public crop recommendation dataset containing nitrogen, phosphorus, potassium, temperature, humidity, pH, rainfall, and crop labels. A Decision Tree Classifier recommends crop type, a KNN-based segmentation component classifies soil zones, and a Linear Regression model predicts a quantitative

Methodology

yield. All models are serialized with joblib and exposed through a Tkinter graphical interface with embedded matplotlib visualizations. The dataset was loaded from 'data/crop_recommendation.csv' and processed through

agricultural machine learning. The inputs were coerced to numeric types, missing values were imputed. The Decision Tree Classifier was trained using the seven numeric input variables to predict the crop label. This model was selected because its feature importance vector supports transparent explanation. The soil segmentation module derives reference clusters using K-Means and trains a KNN classifier to

Results and Discussion

segment the data into three soil zones. The Linear Regression model predicts the yield index. The system achieved 0.9250 accuracy, 0.9142 macro precision, and 0.9250 macro

recall. These results indicate strong crop-label prediction on the selected dataset. The KNN soil segmentation workflow produced a silhouette score of 0.2929, suggesting moderate cluster separation. The cluster scatter plot visualizes soil zone separation using scaled nutrient

Industrial Application

dimensions. The residual plot supports evaluation of the regression module. A key limitation is that the engineered yield index. The residual plot indicates whether regression errors are centered near zero and whether systematic bias remains.

Reports and simulation results to actionable recommendations. The crop classifier can guide crop

Research Extension

selection, the soil segmentation module can support variable-rate fertilizer and irrigation zones, and the yield model can provide a preliminary productivity indicator. With IoT sensors and real-time data, the

architecture could be extended into a deployable advisory dashboard for regional agronomists and

Conclusion

imagery fusion, where vegetation indices, surface temperature, and regional rainfall forecasts could be

combined with tabular soil features using ensemble models or deep learning.

The project demonstrates integration of classification, clustering-oriented segmentation, and regression into a single agricultural intelligence pipeline. It includes modular preprocessing, serialized models, quantitative metrics, visualization artifacts, and a Tkinter interface. The assembled system satisfies the core OEL goal of binding multiple AI components into an executable decision-support application while maintaining clear documentation and reproducible training.